CS3244 Exam 2: Part 1

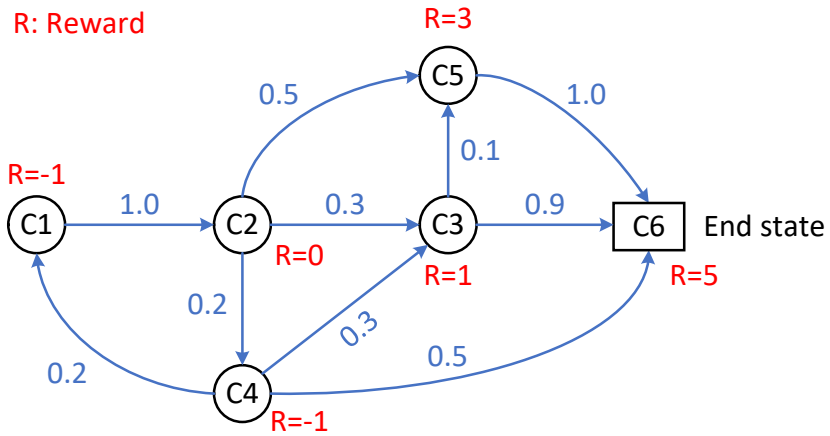
9 November 2020

Please do not turn to the next page until you are told to do so by your proctor.

- This exam part is worth **25** marks out of a **75** mark total for all three parts.
- This exam part is estimated to take you about **25** minutes to complete.
- This exam part has a total of **8** questions.
- This exam part contains only multiple choice questions (MCQs) and multiple response questions (MRQ). Please key these into the assessment system. Please be careful in keying your answers to the corresponding survey item.
- You can visit `http://www.comp.nus.edu.sg/~cs3244/2010/e2.part1.html` to reach the entry form for this survey if you lose your browser window.
- Do remember that you will need to key in and submit your answers to the according assessment system as designated by your proctor or by exam central.

[Questions 1–2] Consider the reward process below, where $\gamma = 0.5$. Suppose we start from C1 and C6 is the terminal state with no self loop.

1. (MCQ; 4 marks) Which of the following episodes yields the highest reward value?



(a) C1 $\rightarrow$ C2 $\rightarrow$ C3 $\rightarrow$ C6.
(b) C1 $\rightarrow$ C2 $\rightarrow$ C4 $\rightarrow$ C6.
(c) C1 $\rightarrow$ C2 $\rightarrow$ C4 $\rightarrow$ C3 $\rightarrow$ C6.
(d) C1 $\rightarrow$ C2 $\rightarrow$ C3 $\rightarrow$ C5 $\rightarrow$ C6.
(e) There is a tie for the highest reward episode among these choices.

[Questions 2–3] In lecture, we have seen a soft order constraint of the form $\boldsymbol{\theta}^\top \boldsymbol{\theta} < C$. For some problems, there exists a more general soft order constraint of the form $\boldsymbol{\theta}^\top \boldsymbol{M}^\top \boldsymbol{M}\boldsymbol{\theta} < C$ which captures the relationship among the θ_i. Let us fix the constant C and $\boldsymbol{\theta} = [\theta_1, \ldots, \theta_n]^\top_{1\times n}$

2. (MCQ; 4 marks) Define M such that we obtain the constraint $\sum \theta_i^2 < 4C$.

(a) $[1, 1, \ldots, 1]^\top_{n\times 1}/4$.
(b) $\boldsymbol{I}_{n\times n}/4$.
(c) $\boldsymbol{I}_{n\times n}/8$.
(d) $[1, 1, \ldots, 1]_{1\times n}/4$.
(e) None of the above.

3. (MCQ; 4 marks) Define M such that we obtain the constraint $(\sum \theta_i)^2 < C$.

(a) $[1, 1, \ldots, 1]^\top_{n\times 1}$.
(b) $\boldsymbol{I}_{n\times n}/4$.
(c) $[1, 1, \ldots, 1]_{1\times n}$.
(d) $\boldsymbol{I}_{n\times n}$.
(e) None of the above.

4. (MCQ; 2 marks) Which of the following episodes yields the highest reward value, when γ is changed to 1.0? (These are the same episodes as above, but perhaps presented in a different order)

   (a) C1 $\rightarrow$ C2 $\rightarrow$ C3 $\rightarrow$ C6.
   (b) C1 $\rightarrow$ C2 $\rightarrow$ C4 $\rightarrow$ C6.
   (c) C1 $\rightarrow$ C2 $\rightarrow$ C3 $\rightarrow$ C5 $\rightarrow$ C6.
   (d) C1 $\rightarrow$ C2 $\rightarrow$ C4 $\rightarrow$ C3 $\rightarrow$ C6.
   (e) There is a tie for the highest reward episode among these choices.

5. (MCQ; 2 marks) We stated that validation generally produces an optimistic estimate of J_{test}. Why?

   (a) Because possibly many parameters are tested in the validation process.
   (b) Because possibly many values of parameters are tested in the validation process.
   (c) Because we choose the model based on their performance.

6. (MRQ; 3 marks) Which of the following statements are true regarding the ethics of data sharing?

   (a) Transparency around the collection of data and how this collected data will be used.
   (b) Ensuring the quality of collected data; and holding sources accountable for low-quality or actively misleading data.
   (c) Clear provenance of data so that data scientists are always aware of where their datasets come from.
   (d) Equitable and ethical access to data once it's collected.

7. (MCQ; 3 marks) A Markov Reward Process

   (a) Will not have a bounded solution when there exist at least one cycle that yields negative net reward.
   (b) Can be modeled as a Markov Decision Process without loss of fidelity.
   (c) Can be flattened to a Markov Chain without loss of fidelity.

8. (MRQ with 4 options; 3 marks) Choose all options that makes the statement true.

   In general, five fold cross validation _____, compared with ten fold cross validation.

   (a) Takes twice the time to complete
   (b) Takes longer to complete
   (c) Takes shorter to complete
   (d) Takes half the time to complete

This marks the end of this part of the exam.
These is no additional material beyond this point.